

# 1 Cross Norms

**Notation 1.1.** In this section,  $A_1, A_2$  will denote Banach spaces. The norms on these spaces will be denoted by  $\|\cdot\|$  unambiguously as we will distinguish between elements of  $A_1$  and  $A_2$ .

**Notation 1.2.** For Banach spaces  $A_1, A_2$ , we will denote their *algebraic tensor product* by  $A_1 \otimes A_2$ . Its completion in some norm  $\|\cdot\|_\beta$  will be denoted by  $A_1 \otimes_\beta A_2$ .

**Definition 1.3.** A norm  $\|\cdot\|_c$  on  $A_1 \otimes A_2$  is called a *cross norm* if it satisfies  $\|x_1 \otimes x_2\|_c = \|x_1\| \|x_2\|$  for all  $x_1 \in A_1, x_2 \in A_2$ .

*Remark 1.4.* That  $\|\cdot\|_\lambda$  is a cross norm is immediate from an application of the Hahn Banach Theorem.

Suppose

**Definition 1.5.** The *injective cross norm*  $\|\cdot\|_\lambda$  on  $A_1 \otimes A_2$  is defined by

$$\|\xi\| = \sup\left\{\left|\sum_{k=1}^n f(x_{1,k})g(x_{2,k})\right| : \begin{array}{l} \xi = \sum_{k=1}^n x_{1,k} \otimes x_{2,k} = \xi, \\ f \in A_1^*, \|f\| = 1, \\ g \in A_2^*, \|g\| = 1 \end{array}\right\}.$$

*Remark 1.6.* Let  $\|\cdot\|_\beta$  be a cross norm on  $A_1 \otimes A_2$ . Retaining the notation of the previous definition,

$$|f(x_1)| |g(x_2)| \leq \|x_1\| \|x_2\| = \|x_1 \otimes x_2\|_\beta.$$

So, the injective cross norm is the smallest cross norm on  $A_1 \otimes A_2$ .

**Definition 1.7.** The *projective cross norm* on  $A_1 \otimes A_2$  is denoted by  $\|\cdot\|_\gamma$  and defined as

$$\|\xi\| = \inf\left\{\sum_{k=1}^n \|x_{1,k}\| \|x_{2,k}\| : \xi = \sum_{k=1}^n x_{1,k} \otimes x_{2,k}\right\}.$$

*Remark 1.8.* Let  $\|\cdot\|_\beta$  be a cross norm on  $A_1 \otimes A_2$ . For  $\xi$  in  $A_1 \otimes A_2$ , and  $\epsilon > 0$ , there exists a representation  $\xi = \sum_{k=1}^n x_{1,k} \otimes x_{2,k}$  in  $A_1 \otimes A_2$ , such that

$$\|\xi\|_\beta \leq \sum_{k=1}^n \|x_{1,k} \otimes x_{2,k}\|_\beta = \sum_{k=1}^n \|x_{1,k}\| \|x_{2,k}\| \leq \|\xi\|_\gamma + \epsilon.$$

So, the projective norm majorises all cross norms. That it is itself a cross norm is evident from the computation

$$\|x_1\| \|x_2\| = \|x_1 \otimes x_2\|_\gamma \leq \|x_1 \otimes x_2\|_\lambda \leq \|x_1\| \|x_2\|.$$

The same computation reveals that all norms ‘between’ the two norms are cross norms. projective norm

## 1.1 $C^*$ cross-norms

In general, the projective cross-norm is not a  $C^*$  norm [1, Remark 4.4]. So we consider

**Definition 1.9.** The maximal  $C^*$  norm defined by

$$\|x\|_{\max} = \sup\{\|\pi(x)\| : \pi \text{ is a representation of } A_1 \otimes A_2\}$$

The max norm is a cross norm.

*Remark 1.10.* For  $\xi = \sum_{k=1}^n x_{1,k} \otimes x_{2,k}$ , in  $A_1 \otimes A_2$ , we have

$$\|\pi(\xi)\| = \left\| \sum_{k=1}^n \pi(x_{1,k} \otimes x_{2,k}) \right\| \leq \sum_{k=1}^n \|\pi(x_{1,k} \otimes x_{2,k})\| \leq \sum_{k=1}^n \|x_{1,k} \otimes x_{2,k}\|.$$

This shows that  $\|x\|_{\max} \leq \|x\|_{\gamma}$ .

In addition, for any representation  $\pi_1$  of  $A_1$  and  $\pi_2$  of  $A_2$ , we have

$$\begin{aligned} \|\pi_1(x_1)\| \|\pi_2(x_2)\| &= \|\pi_1(x_1)\pi_2(x_2)\| = \|(\pi_1 \otimes \pi_2)(x_1 \otimes x_2)\| \\ &\leq \|x_1 \otimes x_2\|_{\max} \leq \|x_1 \otimes x_2\|_{\gamma} \leq \|x_1\| \|x_2\|. \end{aligned}$$

By choosing faithful representations  $\pi_1$  and  $\pi_2$ , the left hand side and right hand side become equal.

**Definition 1.11.** The minimal  $C^*$  cross-norm  $\|\cdot\|_{\min}$  is defined by

$$\|\xi\|_{\min} = \sup\{\|(\pi_1 \otimes \pi_2)(\xi)\| : \begin{array}{l} \pi_1 \text{ is a representation of } A_1, \\ \pi_2 \text{ is a representation of } A_2 \end{array} \}.$$

It is immediate that  $\|\cdot\|_{\min} \leq \|\cdot\|_{\max} \leq \|\cdot\|_{\gamma}$ .

*Remark 1.12.* To see that  $\|\cdot\|_{\min}$  is a cross norm, we only need to show that  $\|\cdot\|_{\Lambda} \leq \|\cdot\|_{\min}$ . For  $f$  in  $A_1^*$ , there exist a representation  $\pi_1$  of  $A_1$  on a Hilbert Space  $\mathcal{H}_1$  and  $\xi_1, \eta_1$  in  $\mathcal{H}_1$  satisfying  $\|f\| = \|\xi_1\| \|\eta_1\|$  and  $f(a) = \langle \eta_1, \pi_1(a)\xi_1 \rangle$  for all  $a$  in  $A_1$ . Similarly, for  $g$  in  $A_2^*$ , there exist a representation  $\pi_2$  of  $A_2$  on a Hilbert space  $\mathcal{H}_2$  and  $\xi_2, \eta_2$  in  $\mathcal{H}_2$  such that  $g(b) = \langle \eta_2, \pi_2(b)\xi_2 \rangle$  for all  $b$  in  $A_2$ .

and  $\|g\| = \|\xi_2\| \|\eta_2\|$ . This means for  $x = \sum_{k=1}^n x_{1,k} \otimes x_{2,k}$  in  $A_1 \otimes A_2$ , we have

$$\begin{aligned}
& |(f \otimes g)x| \\
&= \left| \sum_{n=1}^k f(x_{1,k})g(x_{2,k}) \right| \\
&\leq \left| \sum_{n=1}^k f(x_{1,k})g(x_{2,k}) \right| \\
&\leq \left| \sum_{k=1}^n \langle \eta_1, \pi_1(x_{1,k})\xi_1 \rangle \langle \xi, \pi_2(x_{2,k})\xi_2 \rangle \right| \\
&= \left| \sum_{k=1}^n \langle \eta_1, \pi_1(x_{1,k})\xi_1 \rangle \langle \xi, \pi_2(x_{2,k})\xi_2 \rangle \right| \\
&= \left| \sum_{n=1}^k \langle \eta_1 \otimes \eta_2, (\pi_1 \otimes \pi_2)(x_{1,k} \otimes x_{2,k})(\xi_1 \otimes \xi_2) \rangle \right| \\
&= |\langle \eta_1 \otimes \eta_2, (\pi_1 \otimes \pi_2)(x)(\xi_1 \otimes \xi_2) \rangle| \\
&\leq \|\eta_1\| \|\eta_2\| \|(\pi_1 \otimes \pi_2)(x)\| \|\xi_1\| \|\xi_2\| \\
&= \|f\| \|g\| \|x\|_{\min}
\end{aligned}$$

In other words,  $\|x\|_{\gamma} \leq \|x\|_{\min}$ .

Thus,  $\|\cdot\|_{\min}$  is a cross norm.

## 1.2 Universal Property

**Theorem 1.13.** *For representations  $\pi_1$ , respectively  $\pi_2$ , of  $A_1$ , respectively  $A_2$ , onto a Hilbert Space  $\mathcal{H}$  such that the images of  $\pi_1$  and  $\pi_2$  commute, there exists a unique representation  $\pi$  of  $A_1 \otimes_{\max} A_2$  on the same space  $\mathcal{H}$  such that*

$$\pi(x_1 \otimes x_2) = \pi_1(x_1)\pi_2(x_2)$$

and the image of  $\pi$  is the  $C^*$ -algebra generated by the images of  $\pi_1$  and  $\pi_2$ .

*Proof.* Define  $\pi(x_1 \otimes x_2) = \pi_1(x_1)\pi_2(x_2)$  for  $x_1$  in  $A_1$ , and  $x_2$  in  $A_2$ . For  $x = \sum_{k=1}^n x_{1,k} \otimes x_{2,k}$  and  $y = \sum_{l=1}^m y_{1,l} \otimes y_{2,l}$  in  $A_1 \otimes A_2$ , we have

$$\begin{aligned}
& (\pi_1(x_{1,k}) \otimes \pi_2(x_{2,k}))(\pi_1(y_{1,l}) \otimes \pi_2(y_{2,l})) \\
&= \pi_1(x_{1,k})\pi_2(x_{2,k})\pi_1(y_{1,l})\pi_2(y_{2,l}) \\
&= \pi_1(x_{1,k})\pi_1(y_{1,l})\pi_2(x_{2,k})\pi_2(y_{2,l}) \\
&= \pi_1(x_{1,k}y_{1,l})\pi_2(x_{2,k}y_{2,l}),
\end{aligned}$$

for every  $k$  in  $\{1, 2, \dots, n\}$  and  $l$  in  $\{1, 2, \dots, m\}$ . Taking linear combinations over  $k, l$  we get

$$\pi(x)\pi(y) = \pi(xy).$$

Also,  $\|(\pi_1 \otimes \pi_2)(\sum_{k=1}^n x_{1,k} \otimes x_{2,k})\| \leq \|\sum_{k=1}^n x_{1,k} \otimes x_{2,k}\|_{\max}$ . So  $\pi$  is norm decreasing. Hence there exists an extension of  $\pi$  to all of  $A_1 \otimes_{\max} A_2$ .

Moreover, any other representation  $\rho$  which satisfies  $\rho(x_1 \otimes x_2) = \pi_1(x_1) \otimes \pi_2(x_2)$  must agree with  $\pi$  on the completion because they agree on a dense subset.  $\square$

*Remark 1.14.* This shows that  $A_1 \otimes_{\max} A_2$  is embedded inside  $A_1 \otimes_{\beta} A_2$  for every  $C^*$  cross-norm  $\beta$ .

**Theorem 1.15.** *Let  $A_1, A_2$  be  $C^*$ -algebras and Denote  $A_1 \otimes_{\min} A_2$  by  $A_{\min}$  and  $A_1 \otimes_{\max} A_2$  by  $A_{\max}$ . Then*

1. *If  $\omega_1 \in (A_1^*)_+$ , and  $\omega_2 \in (A_2^*)_+$ , then  $\omega = \omega_1 \otimes \omega_2$  is in  $((A_{\max}^*)_+)$  and induces the representation  $(\pi, \mathcal{H}, \xi)$  which is the tensor product of the representations  $(\pi_1, \mathcal{H}_1, \xi_1)$  and  $(\pi_2, \mathcal{H}_2, \xi_2)$ .*

2.

$$\|x\|_{\min} = \sup \left\{ \frac{(\omega_1 \otimes \omega_2)(a^* x a)^{1/2}}{(\omega_1 \otimes \omega_2)(a^* a)^{1/2}} : \begin{array}{l} \omega_1 \in (A_1^*)_+, \\ \omega_2 \in (A_2^*)_+, \\ a \in A_{\min} \end{array} \right\}$$

*Proof.* Define  $\mathcal{H} = \mathcal{H}_1 \otimes \mathcal{H}_2$ ,  $\xi = \xi_1 \otimes \xi_2$  and  $\pi = \pi_1 \otimes \pi_2$ . Then,

$$\begin{aligned} & \langle \xi, \pi(x_1 \otimes x_2) \xi \rangle \\ &= \langle \xi_1 \otimes \xi_2, (\pi_1(x_1) \otimes \pi_2(x_2))(\xi_1 \otimes \xi_2) \rangle \\ &= \omega_1(x_1) \omega_2(x_2) \\ &= (\omega_1 \otimes \omega_2)(x_1 \otimes x_2). \end{aligned}$$

Taking finite linear combinations, we get that  $\omega(x) = \langle \xi, \pi(x) \xi \rangle$  for all  $x$  in the algebraic tensor product  $A_1 \otimes A_2$ . This shows that  $\omega$  is positive in  $A_{\max}$ . Also,  $\pi_1(A_1)\xi_1$  is dense in  $\mathcal{H}_1$ ,  $\pi_2(A_2)\xi_2$  is dense in  $\mathcal{H}_2$ . So their tensor product  $(\pi_1(A_1)\xi_1) \otimes (\pi_2(A_2)\xi_2) = \pi(A_1 \otimes A_2)\xi$  is dense in  $\mathcal{H}$ . Thus,  $\xi$  is cyclic under  $\pi$ .

For the second assertion, pick  $x$  in  $A_{\min}$ . Given representations  $\pi_1$ , respectively  $\pi_2$ , of  $A_1$ , respectively  $A_2$ , we can express them as direct sums of irreducible representations as  $\pi_1 = \oplus_i \pi_{1,i}$  and  $\pi_2 = \oplus_j \pi_{2,j}$ . Then

$$\|(\pi_1 \otimes \pi_2)(x)\| = \sup_{i,j} \{(\pi_{1,i} \otimes \pi_{2,j})(x)\}.$$

So that

$$\|x\|_{\min} = \sup \|(\pi_1 \otimes \pi_2)(x)\|,$$

where the sup is taken over cyclic representations only. **proof is incomplete.** Let  $\xi_i$  be the cyclic vector for  $\pi_i$ ,  $i = 1, 2$ . Replacing  $\pi_1$ , respectively  $\pi_2$  with

$\omega_1, \in (A_1^*)_+$ , respectively  $\omega_2 \in (A_2^*)_+$ , we get

$$\begin{aligned}
& \|(\pi_1 \otimes \pi_2)(x)\| \\
&= \|(\pi_1 \otimes \pi_2)(x)^*(\pi_1 \otimes \pi_2)(x)\|^{1/2} \\
&= \sup\{\langle \eta, (\pi_1 \otimes \pi_2)(x^*x)\eta \rangle : \eta \neq 0\} \\
&= \sup\left\{\frac{\langle (\pi_1 \otimes \pi_2)(a)\xi, (\pi_1 \otimes \pi_2)(x^*x)(\pi_1 \otimes \pi_2)(a)\xi \rangle}{\langle (\pi_1 \otimes \pi_2)(a)\xi, (\pi_1 \otimes \pi_2)(a)\xi \rangle}\right\} \\
&= \sup\left\{\frac{\langle \xi, (\pi_1 \otimes \pi_2)(a^*x^*xa)\xi \rangle}{\langle \xi, (\pi_1 \otimes \pi_2)(a^*a)\xi \rangle} : a \neq 0, a \in A_1 \otimes A_2\right\} \\
&= \sup\left\{\frac{(\omega_1 \otimes \omega_2)(a^*x^*xa)}{(\omega_1 \otimes \omega_2)(x^*x)} : a \in A_1 \otimes A_2\right\}
\end{aligned}$$

□

tensor products of ucp maps functoriality and universal property

## References

- [1] Gilles Pisier. *Tensor Products of C\*-Algebras and Operator Spaces: The Connes–Kirchberg Problem*. London Mathematical Society Student Texts. Cambridge University Press, 2020.